

COSC 594/650

Gradient Descent for Strongly Convex Objectives

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Convexity vs strong convexity

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (differentiable)

■ CONVEX FUNCTION

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

★ $\nabla^2 f(x) \succeq 0$ – PSD

■ STRONGLY CONVEX FUNCTION

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2$$

★ $\nabla^2 f(x) \succeq \mu I$ – PD

★ $\mu I \preceq \nabla^2 f(x) \preceq LI$ – PD and Lipschitz gradient (L)

★ $\kappa = \frac{L}{\mu}$ – condition number

Convergence analysis for strongly convex functions

Gradient descent: $x_{k+1} = x_k - \alpha \nabla f(x_k)$, $k = 0, 1, 2, \dots$

■ DESCENT LEMMA

$$f(x_{k+1}) \leq f(x_k) - \frac{2\alpha - \alpha^2 L}{2} \|\nabla f(x_k)\|^2, \quad \alpha < \frac{2}{L}$$

■ STRONG CONVEXITY

$$\begin{aligned} f(x^*) &\geq f(x_k) + \langle \nabla f(x_k), x^* - x_k \rangle + \frac{\mu}{2} \|x_k - x^*\|^2 \\ &\geq f(x_k) - \frac{1}{2\mu} \|\nabla f(x_k)\|^2 \end{aligned}$$

$$\implies f(x_{k+1}) - f(x^*) \leq f(x_k) - f(x^*) - \mu(2\alpha - \alpha^2 L)(f(x_k) - f(x^*)), \quad \alpha = \frac{1}{L}$$

$$\iff f(x_k) - f(x^*) \leq \left(1 - \frac{1}{\kappa}\right)^k (f(x_0) - f(x^*))$$

Iteration complexity for strongly convex functions

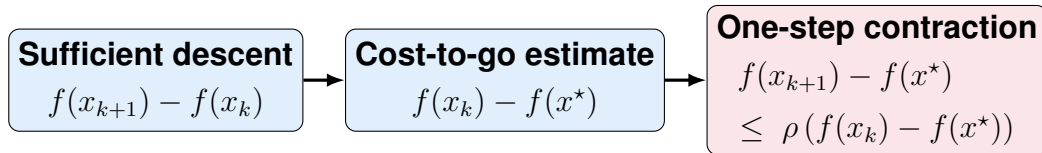
$$e(x_k) \leq \left(1 - \frac{1}{\kappa}\right)^k e(x_0) \leq \epsilon$$

$$\implies k \geq \kappa \log \frac{e(x_0)}{\epsilon}$$

Iteration complexity: $\mathcal{O}\left(\kappa \log \frac{e(x_0)}{\epsilon}\right)$

The same as the quadratic case

Proof machinery



Iterate the contraction:

$$e_{k+1} \leq \rho e_k \implies e_k \leq \rho^k e_0$$

Linear regression

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	5	4	5	3

$$y_i = \theta^\top x_i \longrightarrow$$

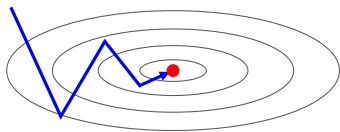
Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_n	4

$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & \|X\theta - y\|^2 \\ \text{subject to} & \theta \in \mathbb{R}^n \end{array}$$

Ill-conditioned $X^\top X$:

Columns of X are nearly duplicates: $\lambda_{\min}(X^\top X) \approx 0$

GD zig-zags slowly toward θ^* (Large κ)



Questions?

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